RUNSHI TANG

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EDUCATION

Ph.D. in Statistics 2021–2026
Department of Statistics, University of Wisconsin–Madison
Advisor: Prof. Anru Zhang; Co-advisor: Prof. Richard Chappell

M.S. in Statistics, Data Science Track 2019–2021
Department of Statistics, University of Wisconsin–Madison
Academic Excellence Award, 2019–2020

B.S. in Mathematics 2016–2020
School of Mathematics, Shandong University
Academic Excellence Award, 2016–2017 and 2017–2018
Exchange student, University of Wisconsin–Madison, 2019–2020

ACADEMIC APPOINTMENTS

Founder's Postdoctoral Research Scientist 2026–present
Department of Statistics, Columbia University
Postdoctoral Mentor: Prof. Genevera Allen

Adjunct Assistant Professor of Statistics 2026–present
Department of Statistics, Columbia University

PAPERS

(†: equal contribution)

1. **Tang, R.**, Yuan, M., & Zhang, A. R. (2025). Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model. *Journal of the Royal Statistical Society, Series B*, 87, 232–255.
2. **Tang, R.**, Kolda, T. G., & Zhang, A. R. (2026). Tensor Decomposition with Unaligned Observations. *SIAM Journal on Matrix Analysis and Applications*, 47(1), 107–131.
3. Towe, S. L., **Tang, R.**, Gibson, M. J., Zhang, A. R., & Meade, C. S. (2023). Longitudinal Changes in Neurocognitive Performance Related to Drug Use Intensity in a Sample of Persons with and without HIV Who Use Illicit Stimulants. *Drug and Alcohol Dependence*, 251, 110923.
4. Zhang, A. R.†, Bell, R.†, An, C.†, **Tang, R.**†, Hall, S., Chan, C., Al-Khalil, K., & Meade, C. (2023). Cocaine Use Prediction with Tensor-based Machine Learning on Multimodal MRI Connectome Data. *Neural Computation*, 36, 107–127.
5. Ko, A., Berger, L., Brown, D., Pac, J., & **Tang, R.** (2025). Unemployment Insurance Generosity and Child Protective Services Involvement. *Child Abuse & Neglect*, 167, 107583.
6. McCann, J. R., Yang, C., Bihlmeyer, N., **Tang, R.**, et al. (2026+). Branched Chain Amino Acid Metabolism and Microbiome in Adolescents with Obesity During Weight Loss Therapy. *The Journal of Clinical Investigation*, 136 (14).
7. **Tang, R.**, Chhor, J., Klopp, O., & Zhang, A. R. (2026+). Revisit CP Tensor Decomposition: Statistical Optimality and Fast Convergence. Submitted to *The Annals of Statistics*, major revision submitted.
8. **Tang, R.**†, Zheng, R.†, Zhang, A. R., & Priebe, C. (2026+). A Functional Tensor Model for Dynamic Multilayer Networks with Common Invariant Subspaces and RKHS Estimation. *Journal of Computational and Graphical Statistics*, accepted.

9. **Tang, R.**, Chhor, J., Klopp, O., Tsybakov, A., & Zhang, A. R. (2026+). Optimal Estimation of Discrete Multiview Distributions under Heteroskedastic Multinomial Sampling. Submitted to *The Annals of Statistics*
10. **Tang, R.**, Han, Y., & Zhang, A. R. (2026+). Detection is harder than estimation in certain regimes: Inference for moment and cumulant tensors. Submitted to *SIAM Journal on Mathematics of Data Science*.
11. **Tang, R.**, Han, Y., & Zhang, A. R. (2026+). From Reverse Detection–Estimation Gaps to Computational Lower Bounds for Norm Approximation. Submitted to *IEEE Transactions on Information Theory*.
12. Pac, J., Berger, L., Brown, D., & **Tang, R.** (2026+). Does Public Program Participation Impact Child Protective Services Involvement? Quasi-Experimental Evidence from the Staggered eWIC EBT Rollout. Working paper.
13. Pac, J., Berger, L., Durrance, C., Nobles, J., & **Tang, R.** (2026+). Foster Care in the First Year of Life: Effects on Health and Safety. Working paper.
14. Pac, J., Berger, L., Cancian, M., Durrance, C., Nobles, J., & **Tang, R.** (2026+). Collateral Harm or Help? The Effect of Foster Care on Mothers. Submitted to American Economic Review.
15. Pac, J., Pilarz, A. R., Slopen, M., & **Tang, R.** (2026+). Collective Impacts: PFML Interacts with State Policies to Affect Maternal Labor Force Participation. Working paper.
16. **Tang, R.**, Zhang, A. R., Han, Y., & Wu, W.-B. (2026+). Bandable Cumulant Tensors: Optimal Estimation and Applications in Non-Gaussian Data Modeling. Submitted to *Journal of the Royal Statistical Society, Series B*.
17. **Tang, R.**, & Allen, G. I. (2026+). RECAP: Representation-Based Calibration with Amortized Pooling for Frequentist Simulation-Based Inference. Submitted to *Journal of the American Statistical Association*.

CONFERENCE/WORKSHOP ABSTRACTS

1. McCann, J., Bihlmeyer, N., Yang, C., Truong, T., **Tang, R.** et al. (2024). Unique Age and Sex Dependent Serum Metabolite Signature in Teens With Obesity. *Abstracts from the 42nd annual meeting of the Obesity Society at Obesity Week November 3-6, 2024*
2. Pac, J., Berger, L., Durrance, C., Nobles, J., & **Tang, R.** (2025) Foster care in the First Year of Life: Effects on Health and Safety. *2024 Association for Public Policy Analysis and Management* and *2025 Atlanta Workshop on Public Policy and Child Well-being*
3. Pac, J., Pilarz, A., Slopen, M., Brook, S., & **Tang, R.** (2026) Collective impacts: Paid Family and Medical Leave interacts with state policies to predict maternal employment *2026 Work and Family Researchers Network Conference*

AWARDS

1. Honorable Mention, 2024 Student Paper Competition, ASA Statistical Learning and Data Science Section, "Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model."
2. ASA Student and Early Career Travel Fund.
3. Student Research Grants Competition, University of Wisconsin–Madison.
4. Travel Scholarship, Sandra Rosenbaum School of Social Work and Institute for Research on Poverty, University of Wisconsin–Madison.
5. 2025 IMS Hannan Graduate Student Travel Award.
6. NSF Early-Career Travel Funding for the 2026 Conference of the International Linear Algebra Society (ILAS) Conference.
7. Paper Award at the 2026 SLDS Conference, "Detection is harder than estimation in certain regimes: Inference for moment and cumulant tensors."
8. Financial-Support Award, Statistics Beyond Euclid: Functional Data, Random Objects and AI, UC Davis, 2026.

TALKS AND PRESENTATIONS

1. The 18th annual CFAR Fall Scientific Retreat, "Longitudinal changes in neurocognitive performance related to drug use intensity in a sample of persons with and without HIV who use illicit stimulants. "

2. The 37th New England Statistics Symposium, "Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model."
3. 2024 Joint Statistical Meetings, "Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model."
4. IMSI Workshop on Statistics Meets Tensors: Methodology, Theory, and Applications, "Revisit CP Tensor Decomposition: Statistical Optimality and Fast Convergence."
5. 2025 Joint Statistical Meetings, "Revisit CP Tensor Decomposition: Statistical Optimality and Fast Convergence."
6. 2026 Conference of the International Linear Algebra Society, "Revisit CP Tensor Decomposition: Statistical Optimality and Fast Convergence."
7. 2026 Conference on Statistical Learning and Data Science: Inference and Intelligence, invited presentation, "Statistical and Computational Limits for Cumulant Tensor Inference: A Regime in Which Efficient Detection Is Harder Than Efficient Estimation. "
8. 2026 Statistical Foundations of Latent Structure and Representation Learning (Columbia University), poster presentation, "RECAP: Representation-Based Calibration with Amortized Pooling for Frequentist Simulation-Based Inference."
9. 2026 Statistics Beyond Euclid: Functional Data, Random Objects and AI (UC Davis), poster presentation, "RECAP: Representation-Based Calibration with Amortized Pooling for Frequentist Simulation-Based Inference."
10. SIAM Conference on Discrete Mathematics, invited presentation, "Statistical and Computational Limits for Cumulant Tensor Inference: A Regime in Which Efficient Detection Is Harder Than Efficient Estimation. "

EXPERIENCE

Project Assistant Sep 2022 - May 2026
Institute for Research on Poverty, UW-Madison *Madison, WI*

- Supervised by Prof. Jessica Pac and Prof. Lawrence Berger (School of Social Science & Institute for Research on Poverty, UW-Madison)
- Studied the relationship between child well-being and various social policies.
- Part of our results were presented by Prof. Lawrence Berger in the 2023 International Society for the Prevention of Child Abuse & Neglect Congress

Academia Internship Summer, 2022
School of Medicine, Duke University *Durham, NC*

- Supervised by Prof. Anru Zhang (Department of Biostatistics & Bioinformatics) and Prof. Christina Meade (Department of Psychiatry and Behavioral Sciences)
- Participated in two projects: longitudinal study on HIV contributes to neurocognitive impairments in persons who use stimulants and applying the novel tensor-based method on MRI data to predict cocaine use
- Presented a poster at the 18th annual CFAR Fall Scientific Retreat

Teaching Assistant Sep 2021 - May 2022
Department of Statistics, UW-Madison *Madison, WI*

- Fall 2021: STAT 324, Introductory Applied Statistics for Engineers
- Spring 2022: STAT 324, Introductory Applied Statistics for Engineers

Project Assistant Sep 2020 - Aug. 2021
Department of Statistics, UW-Madison *Madison, WI*

- Supervised by Prof. Awad Hanna (Department of Civil & Environmental Engineering, UW-Madison) and Prof. Wei-Yin Loh (Department of Statistics, UW-Madison)
- Processed and analyzed data for a research project on construction material transportation.

Actuarial Internship Summer, 2018
Taishan Property Insurance Co., Ltd. *Jinan, Shandong Province, China*

- Work for Actuarial Dept.; Learn structure of insurance company and knowledge of Property Insurance and making Actuarial Report; Help make observation report on opponents.

RESEARCH INTERESTS

- High-dimensional statistics, spiked covariance models, tensor decomposition, matrix analysis, and optimization.
- Applications of statistical theory to social science and clinical science.